

Module 12 — CI/CD and Slim CI

Tier: ● Working Effectively · Duration: 90 min · Prerequisites: Module 11

Why this module exists: Every model you merge reaches production. Without CI, a broken Silver model or a missing `grain` comment ships silently, breaks a Power BI report, and wakes someone up at 3am. This module explains how the CI/CD pipeline protects production, why Slim CI matters for cost, and how the GitHub Actions workflow you already use every day actually works.

Agenda

Time	Duration	Topic	Learning Goal	Mode	Participant Activity	Materials	Trainer Notes	Checkpoint
00:00	10 min	Recap Module 11	Confirm advanced model selection mental model	Q&A	Answer from memory	—	Ask all 4 prep questions cold. Probe <code>dbt build --select +fct_deal</code> specifically — it leads directly into the CI <code>--select</code> discussion.	All 4 correct
00:10	10 min	What CI/CD means in dbt	Understand the full dev → PR → CI → merge → PROD loop	Present	Annotate diagram	This doc	Walk the Mermaid diagram. Make the core point explicit: broken models never reach production because CI intercepts them first.	"Where does dbt build run on a PR — in PROD or in a separate CI schema?"
00:20	15 min	PR workflow	Know exactly what runs on PR open vs. on merge to main	Present + live demo	Compare the two trigger blocks	GitHub Actions file	Show the actual workflow file in GitHub. Point out the <code>on: pull_request</code> vs <code>on: push</code> triggers.	"Name two commands that run on merge but not on PR open."
00:35	15 min	Slim CI	Understand <code>state:modified+</code> , <code>manifest.json</code> , and <code>--defer</code>	Present	Sketch the dependency graph	Whiteboard	Use numbers: 18 models × 3 min = 54 min full run vs. 2 modified	"What does <code>state:modified+select?</code> "

Time	Duration	Topic	Learning Goal	Mode	Participant Activity	Materials	Trainer Notes	Checkpoint
							models + 4 downstream = 6 models × 3 min = 18 min. Cost argument makes it concrete.	
00:50	15 min	GitHub Actions walkthrough	Read and understand a real <code>.github/workflows/dbt_ci.yml</code>	Live demo	Follow along, annotate	YAML file	Walk line by line. Cover triggers, secrets, steps. Show what the GitHub Actions UI looks like when a test fails.	"Which step downloads the prod manifest?"
01:05	5 min	Env vars and targets	Know how <code>profiles.yml</code> uses env vars and what <code>target</code> means	Present	—	This doc	Brief. One key point: CI uses a separate target with a CI-specific schema so it never writes to PROD.	"What is the name of the CI target in <code>profiles.yml</code> ?"
01:10	5 min	Documentation quality gate	Know what makes a PR fail for documentation reasons	Present	—	This doc	Emphasise: this is the one enforcement that catches models before they reach production review.	"Name two things that will fail a PR review."
01:15	25 min	Exercise	Debug a broken CI config; write a correct <code>--select</code> command; explain <code>--defer</code>	Practice	Solo + pair-check	Exercise below	Circulate. Most common mistake on Task 1: fixing the <code>--state</code> path but missing the hardcoded secret.	Exercise complete, 3 bugs identified
01:40	10 min	Debrief + prep questions	Consolidate and preview Module 13	Debrief	Verbal	Whiteboard	Ask: "If CI costs \$40/month today and your project doubles in model count,	—

Time	Duration	Topic	Learning Goal	Mode	Participant Activity	Materials	Trainer Notes	Checkpoint
							what does that do to CI cost without Slim CI? Then introduce Module 13 prep questions.	

Content

Part A — What CI/CD Means in dbt

CI/CD is not just "automated tests." It is a set of gates that prevent broken or undocumented code from reaching production.

The full workflow:

```
flowchart LR
    DEV["Developer\npushes branch"]
    PR["Pull Request\nopened"]
    CI["GitHub Actions\ndbt build in CI schema\n(DEV warehouse)"]
    FAIL["❌ PR blocked\nTests failed"]
    REVIEW["Code review\napproved"]
    MERGE["Merge to main"]
    CD["GitHub Actions\nfull dbt build\n+ docs generate\n+ upload to S3"]
    PROD["✅ Production\nupdated"]

    DEV --> PR --> CI
    CI -- "tests pass" --> REVIEW --> MERGE --> CD --> PROD
    CI -- "tests fail" --> FAIL

    classDef passing fill:#ecfdf5,stroke:#16a34a,color:#065f46
    classDef failing fill:#fef2f2,stroke:#dc2626,color:#7f1d1d
    classDef neutral fill:#f8fafc,stroke:#94a3b8,color:#334155
    class PROD,REVIEW,MERGE,CD passing
    class FAIL failing
    class DEV,PR,CI neutral
```

The CI schema (`SILVER_CI` , `GOLD_CI`) is separate from your dev schema and from PROD. CI builds models there, runs tests against them, and discards the schema after the run. Nothing from CI pollutes production data.

The guarantee: If CI passes, the models compiled correctly and every test you've written passed against real data. The reviewer's job is to check logic and documentation — not catch broken SQL.

Part B — PR Workflow

On PR open or push to a PR branch

```
dbt build --select state:modified+ --defer --state ./prod-artifacts
```

- Builds only models whose SQL changed plus their downstream dependents
- Uses production data for unmodified upstream models (`--defer`)
- Runs all tests for the selected models
- If any test fails, the GitHub Actions job exits non-zero and the PR is blocked

On merge to main

```
dbt build                # full project build against PROD
dbt docs generate        # regenerate docs from PROD manifest
aws s3 sync ./target/docs s3://bloomwell-dbt-docs/ # publish
```

The merge pipeline runs the full project — not slim — because it is updating production. It also regenerates and publishes the dbt docs site so the team always has an up-to-date lineage graph.

The CI schema

CI builds into a dedicated schema, not your dev schema and not PROD. The workflow sets:

```
env:
  DBT_TARGET: ci
  DBT_SCHEMA: DBT_CI_${{ github.run_id }}
```

This gives every CI run its own isolated schema. Two PRs can run in parallel without interfering. The schema is deleted after the run.

Why unique schemas matter for parallel PRs: Each CI run uses a unique schema name (`DBT_CI_<run-id>`) so parallel PRs don't overwrite each other's tables or corrupt test results. Without unique schemas, two concurrent PRs would both write to `DBT_CI` — and one PR's tests would run against the other PR's data.

Part C — Slim CI: The Problem Without It

The cost of running everything on every PR

Imagine your project has 18 models. Each model takes an average of 3 minutes to build in Snowflake. A full `dbt build` takes:

```
18 models × 3 min = 54 minutes per PR run
```

Your team opens 10 PRs per week. That is 540 minutes of Snowflake compute per week — just for CI. Most of that compute rebuilds models that did not change.

What `manifest.json` is

Every `dbt build` produces a `manifest.json` in `./target/`. This file contains:

- The full model graph (which models depend on which)
- A hash of each model's compiled SQL

You store the production `manifest.json` in S3 after each successful merge. CI downloads it before running `dbt build`. dbt compares the current SQL hashes against the production hashes to determine what changed.

Note on what gets hashed: dbt hashes the *compiled* SQL after all variables and macros are resolved — not the raw `.sql` file. If you change a variable that affects a model's compiled output, the model is marked as modified even if the raw SQL looks the

same.

First CI run — no production manifest yet: On a project's first CI run, there is no production manifest yet. For the first deployment: run a full `dbt build` without `--select state:modified+` . After it completes and uploads the manifest to S3, every subsequent PR can use Slim CI.

`state:modified+` explained

```
dbt build --select state:modified+ --defer --state ./prod-artifacts
```

Part	Meaning
<code>state:modified</code>	Models whose compiled SQL hash differs from the production manifest
<code>+</code> (trailing)	All downstream dependents of modified models
<code>--state ./prod-artifacts</code>	Path to the directory containing the production <code>manifest.json</code>
<code>--defer</code>	For models not in the selection set, use the production relation instead of rebuilding

Example: You change `fct_deal.sql` . It has two downstream Gold models: `mart_revenue` and `mart_pipeline` . The selection becomes `fct_deal + mart_revenue + mart_pipeline` — 3 models instead of 18. CI time drops from 54 minutes to roughly 9 minutes.

```
18 models × 3 min = 54 min ← without Slim CI
3 models × 3 min = 9 min ← with Slim CI (one fact change)
```

What `--defer` does

Without `--defer` , CI would need to rebuild every upstream model that `fct_deal` depends on — staging models, dimension models — just to have data to test against. With `--defer` , CI skips rebuilding unchanged models and reads directly from the production relations instead.

When `--defer` fails: If an upstream model exists in your branch but was deleted from production, `--defer` will error — dbt can't find the deferred PROD relation. This is intentional: the error tells you that your branch depends on something that no longer exists in production.

flowchart TD

```
STG_CONTACT["stg_hubspot__contacts\n(PROD relation - deferred)"]
STG_DEAL["stg_hubspot__deals\n(PROD relation - deferred)"]
DIM_CONTACT["dim_contact\n(PROD relation - deferred)"]
FCT_DEAL["fct_deal\n← MODIFIED - rebuilt in CI schema"]
MART_REV["mart_revenue\n← Downstream - rebuilt in CI schema"]
MART_PIPE["mart_pipeline\n← Downstream - rebuilt in CI schema"]

STG_CONTACT --> DIM_CONTACT --> FCT_DEAL
STG_DEAL --> FCT_DEAL
FCT_DEAL --> MART_REV
FCT_DEAL --> MART_PIPE
```

```
classDef prod fill:#f0f9ff,stroke:#0369a1,color:#0c4a6e
classDef rebuilt fill:#ecfdf5,stroke:#16a34a,color:#065f46
class STG_CONTACT,STG_DEAL,DIM_CONTACT prod
class FCT_DEAL,MART_REV,MART_PIPE rebuilt
```

Blue = deferred (reads from PROD). Green = rebuilt in CI schema.

Part D — GitHub Actions Workflow

This is the actual workflow file for the project. Walk through it section by section.

```
# .github/workflows/dbt_ci.yml

name: dbt CI/CD

on:
  pull_request:
    branches: [main]
  push:
    branches: [main]

env:
  DBT_PROFILES_DIR: .

jobs:
  # — Job 1: runs on every PR —————
  slim_ci:
    name: Slim CI – changed models only
    runs-on: ubuntu-latest
    if: github.event_name == 'pull_request'

    env:
      DBT_TARGET: ci
      SNOWFLAKE_ACCOUNT: ${ secrets.SNOWFLAKE_ACCOUNT }
      SNOWFLAKE_USER:    ${ secrets.SNOWFLAKE_USER }
      SNOWFLAKE_PASSWORD: ${ secrets.SNOWFLAKE_PASSWORD }
      SNOWFLAKE_DATABASE: ${ secrets.SNOWFLAKE_DATABASE }
      SNOWFLAKE_WAREHOUSE: ${ secrets.SNOWFLAKE_WAREHOUSE }
      SNOWFLAKE_ROLE:    ${ secrets.SNOWFLAKE_ROLE }
      DBT_SCHEMA:        DBT_CI_${ github.run_id }

    steps:
      - name: Checkout
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: '3.11'

      - name: Install dbt
        run: pip install dbt-snowflake==1.8.4

      - name: Download production manifest
        run: |
          mkdir -p ./prod-artifacts
          aws s3 cp s3://bloomwell-dbt-artifacts/prod/manifest.json ./prod-artifacts/manifest.json
        env:
```

```

    AWS_ACCESS_KEY_ID:    ${ secrets.AWS_ACCESS_KEY_ID }
    AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_ACCESS_KEY }
    AWS_REGION:           eu-central-1

- name: dbt deps
  run: dbt deps

- name: dbt build – slim CI
  run: |
    dbt build \
      --select state:modified+ \
      --defer \
      --state ./prod-artifacts \
      --target ci

- name: Drop CI schema
  if: always()
  run: |
    dbt run-operation drop_schema \
      --args "{ 'schema': '$DBT_SCHEMA' }" \
      --target ci

# — Job 2: runs on merge to main —————
deploy:
  name: Deploy to production
  runs-on: ubuntu-latest
  if: github.event_name == 'push' && github.ref == 'refs/heads/main'

  env:
    DBT_TARGET: prod
    SNOWFLAKE_ACCOUNT:  ${ secrets.SNOWFLAKE_ACCOUNT }
    SNOWFLAKE_USER:     ${ secrets.SNOWFLAKE_USER }
    SNOWFLAKE_PASSWORD: ${ secrets.SNOWFLAKE_PASSWORD }
    SNOWFLAKE_DATABASE: ${ secrets.SNOWFLAKE_DATABASE }
    SNOWFLAKE_WAREHOUSE: ${ secrets.SNOWFLAKE_WAREHOUSE }
    SNOWFLAKE_ROLE:     ${ secrets.SNOWFLAKE_ROLE }

  steps:
    - name: Checkout
      uses: actions/checkout@v4

    - name: Set up Python
      uses: actions/setup-python@v5
      with:
        python-version: '3.11'

    - name: Install dbt
      run: pip install dbt-snowflake==1.8.4

    - name: dbt deps
      run: dbt deps

    - name: dbt build – full production run
      run: dbt build --target prod

```

```
- name: Generate dbt docs
  run: dbt docs generate --target prod

- name: Upload manifest to S3 (for next Slim CI run)
  run: |
    aws s3 cp ./target/manifest.json \
      s3://bloomwell-dbt-artifacts/prod/manifest.json
  env:
    AWS_ACCESS_KEY_ID:    ${ secrets.AWS_ACCESS_KEY_ID }
    AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_ACCESS_KEY }
    AWS_REGION:           eu-central-1

- name: Publish dbt docs to S3
  run: |
    aws s3 sync ./target/ \
      s3://bloomwell-dbt-docs/ \
      --include "*.html" --include "*.json" --include "*.js"
  env:
    AWS_ACCESS_KEY_ID:    ${ secrets.AWS_ACCESS_KEY_ID }
    AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_ACCESS_KEY }
    AWS_REGION:           eu-central-1
```

What each section does

Section	What it does
on: pull_request	Triggers slim_ci job on every PR push
on: push to main	Triggers deploy job after merge
if: github.event_name == 'pull_request'	Ensures slim_ci only runs on PRs, not on merge
secrets.*	Reads Snowflake credentials from GitHub repository secrets — never hardcoded

How secrets injection works: GitHub Actions replaces `${ secrets.* }` with the actual secret value at runtime, before the shell command runs. The YAML file always contains the placeholder — secrets are never written to the repository. If a secret is missing from repository settings, the Actions job fails with a clear error.

| `DBT_SCHEMA: DBT_CI_${ github.run_id }` | Unique schema per run — parallel PRs don't collide |

| Download production manifest | Fetches `manifest.json` from S3 so `state:modified+` can compare against it |

| `dbt build --select state:modified+` | Slim CI — only changed models and their downstream dependents |

| Drop CI schema | Cleans up after CI — `if: always()` ensures it runs even if the build fails |

Why `if: always()` on the cleanup step: The `if: always()` flag ensures the schema cleanup step runs even if `dbt build` failed. Without it, a failed CI run would leave behind the throwaway schema, consuming Snowflake storage and potentially confusing future runs.

| `dbt build` (deploy job) | Full production run — all models |

| Upload manifest | Saves the new `manifest.json` to S3 for the next PR's Slim CI to compare against |

| Publish docs | Uploads updated lineage and documentation to the S3 static site |

Part E — Environment Variables and Targets

`profiles.yml` with env vars


```
# profiles.yml (committed to repo – no secrets in this file)
analytics:
  target: dev
  outputs:

    dev:
      type: snowflake
      account:  "{{ env_var('SNOWFLAKE_ACCOUNT') }}"
      user:     "{{ env_var('SNOWFLAKE_USER') }}"
      password: "{{ env_var('SNOWFLAKE_PASSWORD') }}"
      database: "{{ env_var('SNOWFLAKE_DATABASE') }}"
      warehouse: "{{ env_var('SNOWFLAKE_WAREHOUSE') }}"
      role:      "{{ env_var('SNOWFLAKE_ROLE') }}"
      schema:   "TESTING__{{ env_var('DBT_DEVELOPER_NAME', 'dev') }}"

    ci:
      type: snowflake
      account:  "{{ env_var('SNOWFLAKE_ACCOUNT') }}"
      user:     "{{ env_var('SNOWFLAKE_USER') }}"
      password: "{{ env_var('SNOWFLAKE_PASSWORD') }}"
      database: "{{ env_var('SNOWFLAKE_DATABASE') }}"
      warehouse: "{{ env_var('SNOWFLAKE_WAREHOUSE') }}"
      role:      "{{ env_var('SNOWFLAKE_ROLE') }}"
      schema:   "{{ env_var('DBT_SCHEMA', 'DBT_CI') }}"

    prod:
      type: snowflake
      account:  "{{ env_var('SNOWFLAKE_ACCOUNT') }}"
      user:     "{{ env_var('SNOWFLAKE_USER') }}"
      password: "{{ env_var('SNOWFLAKE_PASSWORD') }}"
      database: "{{ env_var('SNOWFLAKE_DATABASE') }}"
      warehouse: "{{ env_var('SNOWFLAKE_WAREHOUSE') }}"
      role:      "{{ env_var('SNOWFLAKE_ROLE') }}"
      schema:   PUBLIC
```

Three targets, three schemas

Target	Who uses it	Schema
dev	Developer on their machine	TESTING__<your-name>
ci	GitHub Actions on PR	DBT_CI_<run-id> (unique per run, discarded after)
prod	GitHub Actions on merge to main	PUBLIC (the real schema)

The key point: CI never writes to `PUBLIC` . It writes to a throwaway schema named after the run ID. Tests run against data in that schema. When CI finishes, the schema is dropped. Production is never touched.

Never hardcode credentials

The `profiles.yml` file is committed to the repository. It must never contain actual passwords. Every credential goes through `env_var()` . Local developers set these in a `.env` file (git-ignored). CI reads them from GitHub repository secrets.

Part F — Documentation Quality Gate

What makes a PR fail

CI enforces documentation requirements as part of the pre-merge review. A PR is blocked if:

1. A Silver or Gold model is missing a `description` in `schema.yml`
2. A model is missing a `grain` comment at the top of the SQL file
3. A `_key` column (surrogate or foreign key) has no `description` in `schema.yml`

These checks are run via the `dbt-sql-reviewer` skill during code review. They are not yet automated in GitHub Actions — the reviewer runs the skill before approving.

Why documentation is a gate, not a suggestion

Documentation degrades in the same way code does — incrementally. One model ships without a description. Two more follow. After six months, half the lineage graph is undocumented. New team members can't tell what `fct_deal` means without reading 400 lines of SQL.

The gate is effective because it is applied at the only moment that creates leverage: before the PR is approved.

What the dbt-sql-reviewer skill checks

The skill reviews every modified SQL and YAML file in the PR for:

- `description` field present and non-empty for every model and column
- `grain:` comment in the SQL (for Silver and Gold models)
- Column-level docs on all `_key` columns
- Correct materialization for the layer
- `on_schema_change: sync_all_columns` for incremental models

Any failed check becomes a review comment. The PR cannot be approved until all checks pass.

Live Demo Script

Goal: Walk through the GitHub Actions workflow file, trigger a test failure, and show what the developer sees.

1. Open the `.github/workflows/dbt_ci.yml` file in GitHub. Walk each section: `on:` triggers, `env:` block, each step.
2. Point out the `secrets.*` references. Show the GitHub repository → Settings → Secrets page (read-only). Note: the values are never visible — only that a secret named `SNOWFLAKE_PASSWORD` exists.
3. Open a PR for a branch that changes `fct_deal.sql`. Show the GitHub Actions UI as the `slim_ci` job runs.
4. Show the `dbt build --select state:modified+ --defer --state ./prod-artifacts` command in the log.
5. Now break a test: add a row to a source that violates a `not_null` test. Push. Show the red X on the PR and the test failure output in the Actions log.
6. Show the "Checks" tab on the PR — GitHub blocks the merge button when the CI job fails.
7. Fix the issue, push again, show the green checkmark, and show the merge button become available.

Exercise (25 min)

Task 1 — Fix the broken CI config

The workflow file below has **three bugs**. Identify each one, explain why it is a problem, and write the fix.

```
name: dbt CI

on:
  pull_request:
    branches: [main]
```

```

jobs:
  slim_ci:
    runs-on: ubuntu-latest

    env:
      DBT_TARGET: ci
      SNOWFLAKE_ACCOUNT: ${ secrets.SNOWFLAKE_ACCOUNT }
      SNOWFLAKE_USER:    ${ secrets.SNOWFLAKE_USER }
      SNOWFLAKE_PASSWORD: "Snowflake_Prod_2024_v3"
      SNOWFLAKE_DATABASE: ${ secrets.SNOWFLAKE_DATABASE }
      SNOWFLAKE_WAREHOUSE: ${ secrets.SNOWFLAKE_WAREHOUSE }
      SNOWFLAKE_ROLE:    ${ secrets.SNOWFLAKE_ROLE }
      DBT_SCHEMA:        DBT_CI_${ github.run_id }

    steps:
      - name: Checkout
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: '3.11'

      - name: Install dbt
        run: pip install dbt-snowflake==1.8.4

      - name: Download production manifest
        run: |
          mkdir -p ./prod-artifacts
          aws s3 cp s3://bloomwell-dbt-artifacts/prod/manifest.json ./prod-artifacts/manifest.json
        env:
          AWS_ACCESS_KEY_ID:    ${ secrets.AWS_ACCESS_KEY_ID }
          AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_ACCESS_KEY }
          AWS_REGION:          eu-central-1

      - name: dbt deps
        run: dbt deps

      - name: dbt build - slim CI
        run: |
          dbt run \
            --select state:modified+ \
            --defer \
            --state ./target \
            --target ci

```

► The three bugs and fixes

Task 2 — Write the CI `--select` command

You are opening a PR that modifies only `fct_deal.sql`. The model graph looks like this:

```
stg_hubspot_deals → fct_deal → mart_revenue
                  → mart_pipeline_summary
```

Write the exact `dbt build` command you would use in CI. You want to:

- Build only the changed model and its downstream dependents
- Use production data for upstream staging models (do not rebuild them)
- Compare against the production manifest stored at `./prod-artifacts`

► Answer

Task 3 — Explain `--defer`

Answer each question in exactly one sentence:

1. In one sentence: what does `--defer` do when dbt encounters an unmodified upstream model?
2. In one sentence: why does this matter for CI compute cost?
3. In one sentence: what would happen without `--defer` if only one Gold mart changed?

► Answer

Reference Material

- [dbt Slim CI docs](#)
- [dbt state: selector](#)
- [dbt --defer flag](#)
- [GitHub Actions: using secrets](#)
- [dbt profiles.yml with env vars](#)
- Internal: `dbt-sql-reviewer` skill — runs the documentation quality gate before PR approval

Prep Questions for Module 13

Module 13 covers Advanced Testing Patterns — custom generic tests, `dbt-expectations`, severity levels, and `store_failures`.

1. What is the difference between a `not_null` test and a `relationships` test in dbt?
2. Where in your project do you define a generic test that can be reused across multiple models?
3. What does `severity: warn` do differently from the default `severity: error`?
4. If a test fails in production during a scheduled run, how does dbt report that failure — and how would you investigate it?

Tier 2 Complete: Working Effectively

You've completed the Working Effectively tier. You can now:

- Build dbt projects with models, seeds, macros, and snapshots
- Use the selection syntax to run exactly what you need, nothing more
- Implement SCD2 history tracking via snapshots or custom macros
- Protect production with a CI/CD pipeline that blocks broken code at the PR gate
- Reduce CI cost with Slim CI so every PR doesn't run your entire project

Tier 3 — Production and Advanced Patterns — covers advanced testing, complex incremental strategies, the scd2_merge macro deep dive, and data governance with model contracts.